Asking Retraction Watch Twice: Retraction Data Revise Their Own Past

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Abstract

Retraction data are increasingly treated as infrastructure: metascience studies count retractions by year and reason, reference managers warn authors who cite retracted work, and retrieval systems filter retracted papers out. Since Crossref opened the Retraction Watch database in 2023, it has been republished daily, and studies cite it as a single source. We replayed 39 snapshots spanning its entire public history (2024-09-02 to 2026-10-16) and asked the same four questions of each, all about retractions dated 2023 or earlier, so that any change is the database revising the past. Over two years the number of such retractions grew 9.1%, the number citing "Paper Mill" grew 80.6%, the median time from publication to retraction rose from 379 to 420 days, and the count citing "Fake Peer Review", 4,654 in the first snapshot, fell to zero when the label was renamed. Almost all of the growth is late cataloguing: 4,482 records added after the first snapshot, 3,921 of them Hindawi retractions. We also join Retraction Watch with OpenAlex, PubMed, and PubMed Central. Of 89,681 papers that at least one registry flags, all eligible registries agree on 51.5%, and 49.3% of the works OpenAlex marks as retracted are retraction notices rather than retracted papers. A blind hand audit, its rules committed publicly before any label, of 200 of the 36,401 papers that only one registry flags estimates that 95.0% (91.5% to 98.6%) were in fact retracted: the registries disagree mostly by omission, not by error. We recommend that studies cite and archive the snapshot they used, and we release the pipeline that rebuilds any snapshot on demand.

1. Introduction

A study that reports "N retractions for paper mill activity through 2023" reports a property of the database on the day it was queried, not a property of the literature. That would not matter if retraction databases only ever appended new notices. They do not. Records are added years after the retraction, reasons are renamed in bulk, and dates and identifiers are corrected.

The question this paper asks is simple: if two researchers ask Retraction Watch the same question about the past on different days, how different are their answers, and why? Answering it needs the database's history, not just its current state. Crossref publishes the Retraction Watch database as a CSV in a public GitLab repository, regenerated daily, and every regeneration is a commit. The history therefore exists, and the maintainers log taxonomy changes, but no study we found reads it as a time series of records. The closest work compares two time points across indexes (Salami et al., 2024b) or tracks Crossref DOIs that lost their retracted flag, and names longitudinal study of data quality as future work (Si et al., 2026).

We make four contributions.

1. A replay of 39 Retraction Watch snapshots covering the whole public history, with every record historized, so the database can be queried as of any snapshot date.

2. A fixed set of questions asked of every snapshot, showing how much and through which mechanisms the answers move (Section 4.1).
3. A four registry comparison (Retraction Watch, OpenAlex, PubMed, PubMed Central) on current data, which replicates the disagreement that earlier studies found across other indexes, and independently measures how often OpenAlex's retraction flag marks the notice rather than the paper (Sections 4.2 and 4.3).
4. A blind hand audit, with its codebook committed before labelling, of the papers that only one registry flags, so that coverage is measured against publishers rather than against the union of the registries (Section 4.4).

2. Related work

Indexes disagree about what is retracted. This is well established. Schneider et al. (2023) compared Crossref, Retraction Watch, Scopus, and Web of Science and found only 3% of 49,924 DOIs flagged by all four. Salami et al. (2024a, corrected 2026) extended the comparison to 11 sources and found at least one disagreement on 92.64% of 83,317 items. Ortega and Delgado-Quirós (2024) compared coverage across seven databases including OpenAlex and PubMed, and Sebo and Sebo (2025) compared Retraction Watch, PubMed, and Web of Science on 878 retracted publications in 131 high impact medical journals. We do not claim disagreement as a finding. Section 4.2 replicates it with OpenAlex and PMC, which the 11 source study did not include.

Notices indexed as retracted papers. Hand checks by the same group found that a retraction index often holds the notice rather than the paper: 59 of 201 DOIs sampled from those on which the four sources disagreed in Schneider et al. (2023), and 661 of 925 sampled Crossref DOIs (71.46%) in Si et al. (2026). For OpenAlex, Hauschke and Nazarovets (2025) reported a different failure: between 2023-12-22 and 2024-03-19 OpenAlex set is_retracted for any work carrying an update relation, such as a correction or an expression of concern, and OpenAlex reported fixing about 2,300 works. A company report (Tesseract Semantics, 2026), not peer reviewed, states that OpenAlex "flags 94.5 per cent of Retraction Watch notice DOIs as retracted". Section 4.3 measures the same thing independently.

Retraction data as a moving target. Si et al. (2026) found that between April 2023 and July 2024 Crossref removed the retracted flag from 208 DOIs, none of which was in fact retracted, and concluded that "retraction metadata are dynamic rather than static". Their study covers Crossref only, and they name longitudinal assessment of data quality as future work. Salami et al. (2024b) compared union lists built in April 2023 and July 2024. Their own 11 source dataset was later corrected because it had included Retraction Watch records whose nature is not a retraction (expressions of concern, corrections, reinstatements) (Salami and McCumber, 2026), which shows how easily the database's structure is misread even by specialists. Retraction Watch documents its taxonomy changes in a dated changelog (Retraction Watch, n.d.). We found no study that compares Retraction Watch releases record by record, or that measures what those changes do to the answers a study gets.

Citation after retraction. Retracted papers keep being cited, mostly without acknowledgement of the retraction (Bar-Ilan and Halevi, 2017; Schneider et al., 2020); Hsiao and Schneider (2021) found that only 722 of 13,252 (5.4%) post retraction citation contexts in the PubMed Central Open Access Subset acknowledged the retraction. Our citation counts (Section 4.5) measure volume at year granularity and do not classify contexts.

3. Data and methods

3.1 Sources

All four sources were fetched on 2026-10-19 from public endpoints.

Source	What was landed
Retraction Watch (Crossref GitLab)	39 snapshots, 2024-09-02 to 2026-10-16; 55,804 records in the first, 73,283 in the last
OpenAlex	136,873 of 136,970 works with `is_retracted: true`
PubMed	35,169 PMIDs with publication type "Retracted Publication"
PubMed Central	metadata for 43,616 articles the other registries mention

The GitLab history begins on 2024-09-02 and held 518 daily snapshots when we listed it on 2026-10-26. We replayed the first snapshot of each month through 2026-09-01 and every snapshot from 2026-09-29 on.

3.2 Snapshot replay

Each snapshot is landed unchanged and replayed in date order. A record is keyed by its Retraction Watch Record ID. When a record's content changes between consecutive snapshots, its current version is closed and a new one opened, so every version carries a validity interval (`valid_from` inclusive, `valid_to` exclusive). The view of the database as of a date d is the set of versions valid at d. Changes that only alter formatting (whitespace, a dropped + prefix on every reason, capitalization) are classified separately from semantic edits, so a reformatting of the whole file does not read as 143,447 edits.

One snapshot, 2026-10-15, arrived in cp1252 rather than UTF-8 and had lost characters in 5,520 records. It is quarantined rather than replayed as edits.

Validation against the maintainers' changelog. Retraction Watch logs its taxonomy and format changes with dates (Retraction Watch, n.d.). Every change it logs inside our window appears in the first replayed snapshot after its logged date, classified the right way:

Logged	Change	First replayed snapshot	What the replay recorded
2024-09-23	"Withdrawal" renamed "Removed"	2024-10-01	relabel, 3,798 records
2025-02-28	about a hundred reason renames	2025-03-03	relabels, 28,223 updated records
2025-03-06	"Unreliable Results" renamed	2025-04-01	relabel, 16,379 records
2025-08-21	"+" prefix removed from reasons	2025-09-02	format only, 65,898 records
2025-11-25	"Concerns/Issues with Peer Review" renamed	2025-12-01	relabel, 10,557 records
2025-12-12	"Fake Peer Review" renamed	2026-01-02	relabel, 10,426 records

The replay therefore recovers the documented changes without being told about them, which is what licenses reading the undocumented ones (late records, DOI and date edits, deletions) as real.

3.3 The same question on every snapshot

We fixed four questions before computing any of them, and report all four:

1. How many retractions are dated 2023 or earlier?
2. How many of those cite the reason "Paper Mill"?
3. How many of those cite the reason "Fake Peer Review"?
4. What is their median time from publication to retraction?

Restricting every question to retractions dated 2023 or earlier means a correct, append only database would give the same answer on every snapshot after early 2024 (allowing for notices dated late in 2023 that take a few weeks to be entered). Any larger change is the database revising the past. Reason matching is case insensitive substring matching on the reason field.

3.4 Cross registry identity and agreement

Papers are matched across sources on DOI, PMID, and PMCID, using one hop of the PMC identifier map. A paper is in the registry if Retraction Watch holds a record of it or any source flags it as retracted. Retraction and withdrawal notices are excluded before matching: OpenAlex works of type retraction, and any work whose DOI or PMID Retraction Watch records as a notice.

A registry is eligible for a paper only if it could have flagged it: PubMed for papers with a PMID, PMC for papers with a linked open copy, OpenAlex and Retraction Watch for every paper. "All eligible agree" means every eligible registry flags the paper.

3.5 Hand audit

Agreement against the union of registries says nothing about whether the union is right. We drew a sample from the 36,401 papers that exactly one registry flags, stratified by that registry with equal allocation: all 22 papers that only PubMed flags, and 89 each from those that only Retraction Watch (20,259) or only OpenAlex (16,120) flags. No paper is flagged by PMC alone. The draw is seeded and reproducible.

Labels follow a codebook (`docs/audit-codebook.md`) committed, with the blank sheet, before any label was entered (commit 772bdad, 2026-10-26). Labels are assigned from the publisher's version of record or the publisher's own notice, never from the four registries under test. The sheet withholds the flagging registry and strips leading "RETRACTED:" and "WITHDRAWN:" markers from titles (86 of 200), which would reveal both the source and the answer. Blinding is partial: 27 papers have no DOI, PMID, or PMCID, and nearly all of those are Retraction Watch records; and 19 of the 22 PubMed only papers have a PMID but no DOI. A labeller could infer the source from which links exist.

Precision per stratum is the share labelled retracted among papers not labelled `cannot_determine`, with Wilson 95% intervals. The frame estimate weights strata by population with a finite population correction, so the PubMed census adds no sampling variance.

4. Results

4.1 Retraction Watch revises its past

Table 1 and Figure 1 give the four answers on the first and last snapshots.

Table 1. The same question, asked twenty five months apart.

Question	2024-09-02	2026-10-16	Change
Retractions dated 2023 or earlier	48,789	53,227	+9.1%
Of those, citing “Paper Mill”	4,868	8,790	+80.6%
Of those, citing “Fake Peer Review”	4,654	0	−100%
Median days, publication to retraction	379	420	+10.8%

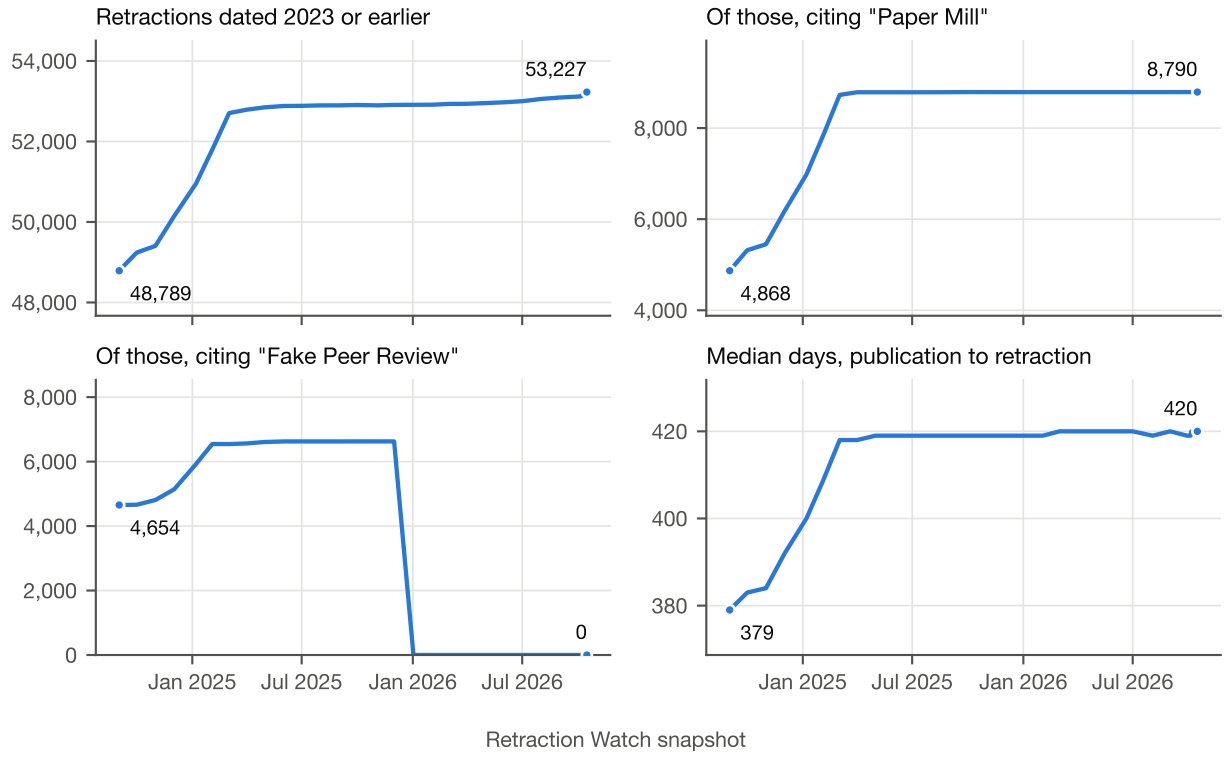


Figure 1. Each panel asks one question of all 39 snapshots.

Three mechanisms account for the movement.

Late cataloguing. 4,482 records with a retraction date in 2023 or earlier were added after the first snapshot. 4,406 of them carry a Record ID above the first snapshot’s highest, so they are new catalogue entries rather than rows the first file happened to omit. 4,315 are dated 2023, and Hindawi accounts for 3,921. Retractions dated 2023 alone went from 9,030 to 13,337: a study of 2023 run in September 2024 would have seen about two thirds of what the database now holds. The late records also explain the other movements: 3,924 of them cite “Paper Mill” (against a net increase of 3,922), and their median lag is 561 days, which pulls the cohort median up. Most arrived between October 2024 and March 2025 (Figure 2), but they were still arriving in October 2026.

Relabelling. The reason taxonomy was renamed in bulk several times. On 2026-01-02 “Fake Peer Review” became “Compromised Peer Review” on 10,426 records; on 2025-04-01 “Unreliable Results” became “Unreliable Results and/or Conclusions” on 16,379. A query written against the old label silently returns zero. Of the 73,334 records ever seen, 50,518 (68.9%) were edited after they first appeared, mostly through these relabels.

Edits and deletions. 3,840 records (5.2%) changed in something other than reasons: 96 changed their original paper DOI, 54 their retraction date, and 27 their nature (for example, from expression of concern to retraction). 54 records were deleted and 3 reappeared. Within the 2023 cohort these are small: 22 records deleted and 22 moved out by a date or nature edit.

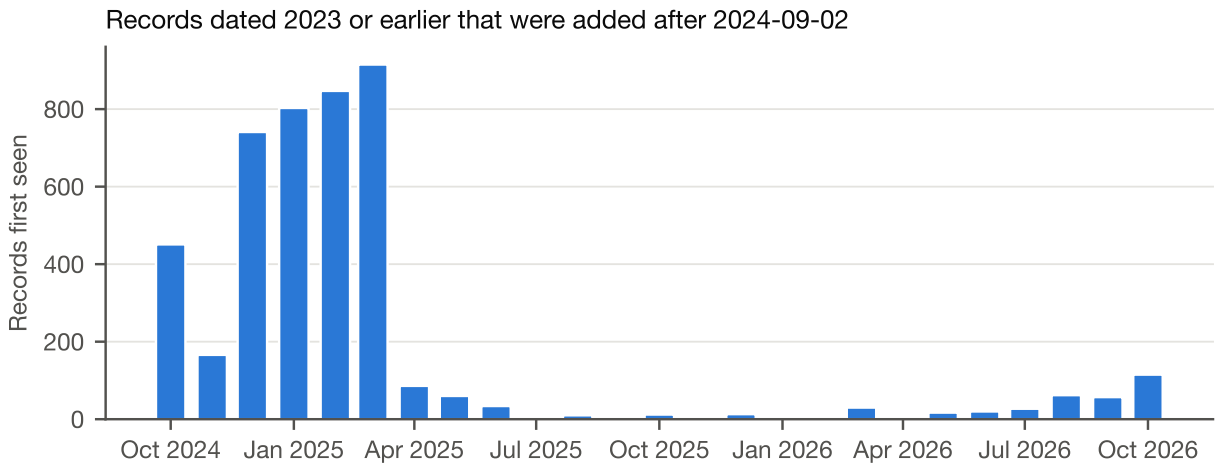


Figure 2. When the late records first appeared.

4.2 The registries disagree

89,681 papers are flagged retracted, or disputed, by at least one registry. All eligible registries agree on 46,225 (51.5%), and exactly one flags 36,401 (40.6%).

Table 2. Each registry against the union. The union is not ground truth; Section 4.4 measures it.

Registry	Eligible papers	Flags	Share of the union
PMC	19,499	18,748	96.1%
PubMed	37,495	33,302	88.8%
OpenAlex	84,162	68,945	81.9%
Retraction Watch	89,681	67,672	75.5%

These are not directly comparable with Salami et al. (2024a), who measure agreement over the union of 11 sources without conditioning on whether a source could have held a paper. The one registry both studies include behaves alike: in Salami et al. (2024a) PubMed had the highest agreement score, indexing as retracted 22,766 (87.69%) of the 25,962 union list items it covers, and here it covers 88.8% of the eligible union. Retraction Watch sits lowest here (75.5%) because the union includes OpenAlex, whose flag reaches many papers no other registry marks; Section 4.4 asks how many of those are real.

4.3 Half of OpenAlex's retracted works are notices

Of the 136,873 OpenAlex works with `is_retracted: true`, 67,544 (49.3%) are retraction notices: 38,161 have type retraction, and 29,383 more carry a DOI or PMID that Retraction Watch records as a notice. Counted as papers, they would nearly double the registry. PMC handles the same distinction well: of the 17,951 notice articles it holds, 187 carry its retracted flag.

From the notice's side the picture is the same. Of the 44,651 Retraction Watch retraction notice DOIs that are not also some paper's DOI, OpenAlex flags 41,460 (92.9%) as retracted. This independently reproduces, within two points, the 94.5% in the Tesseract Semantics (2026) report, on data fetched two months later.

It is not the failure Hauschke and Nazarovets (2025) reported. That one flagged corrected papers and ran for three months in early 2024; this one flags the notice that announces a retraction, is current, and affects nine in ten notices.

4.4 What the single registry flags really are

Of the 200 sampled papers, 183 could be decided from the publisher's page. The other 17 could not: 14 of them are papers only Retraction Watch flags, and 12 of those 14 have no DOI, PMID, or PMCID to follow. Across all 200, the labels were retracted 173 times, notice 6, not_retracted 3, and concern once.

Table 3. Precision of a flag that no other registry shares: the share the publisher actually retracted, among papers that could be decided.

Flagged only by	Papers in frame	Sampled	Decided	Precision (95% CI)	Notices
OpenAlex	16,120	89	86	98.8% (93.7% to 99.8%)	1
PubMed	22	22	22	86.4% (66.7% to 95.2%)	1
Retraction Watch	20,259	89	75	92.0% (83.6% to 96.3%)	4

Weighted by stratum size, with a finite population correction for the PubMed census, 95.0% (91.5% to 98.6%) of the 36,401 single registry flags are real retractions. The registries therefore disagree mostly because one holds a retraction the others miss, not because one flags papers that were never retracted. Treated as a measure of coverage, the union in Table 2 is close to right.

Three cautions. The OpenAlex figure is precision after the notice filter of Section 3.4, not of the raw is_retracted flag, which Section 4.3 shows is half notices. Papers flagged only by OpenAlex may still be in Retraction Watch under a record with no DOI or PMID, which one hop identity resolution cannot match. And the Retraction Watch stratum rests on 75 decided papers: counting all 14 undecided ones as not retracted, the worst case, it would be 69 of 89.

The errors that do exist are instructive. Notices counted as papers make up 6 of the 200, the same failure as Section 4.3 at a smaller scale and found in all three strata.

4.5 Two secondary observations

Publisher batches set yearly medians. Across 67,845 dated notices the median time to retraction is 493 days (interquartile range 156 to 1,074). But one publisher's batch can set a year's median. IEEE conference retractions make up 4,410 of 5,011 notices in 2010 and 4,091 of 4,937 in 2011, with medians of 54 and 34 days; without IEEE those years sit at 548 and 553 days. In 2023, Hindawi's 9,673 notices have a median of 467 days against 954 for everyone else (Figure 3). A yearly trend in retraction speed is largely a trend in which publishers retracted in bulk that year.

Retracted papers keep being cited. Of 44,557 retracted papers whose whole post retraction period falls inside OpenAlex's citation window, 29,024 (65.1%) were cited in at least one year after their retraction year, 217,376 citations in all; the median paper received 1 and the 90th percentile 11. Year granularity undercounts, and these counts do not distinguish citations that acknowledge the retraction from those that do not.

5. Discussion

The headline numbers in Section 4.1 are not errors in Retraction Watch. Late cataloguing is what a curated database does when a publisher retracts eight thousand papers at once, and renaming a reason is what a maintained taxonomy does. The problem is on the side of the studies that use it: a count from this database is a count as of a date, and most studies give neither the date nor the file.

This is the longitudinal study Si et al. (2026) call for, done for Retraction Watch rather than Crossref, and it agrees with their conclusion that researchers "may obtain different results about retraction status depending on when the

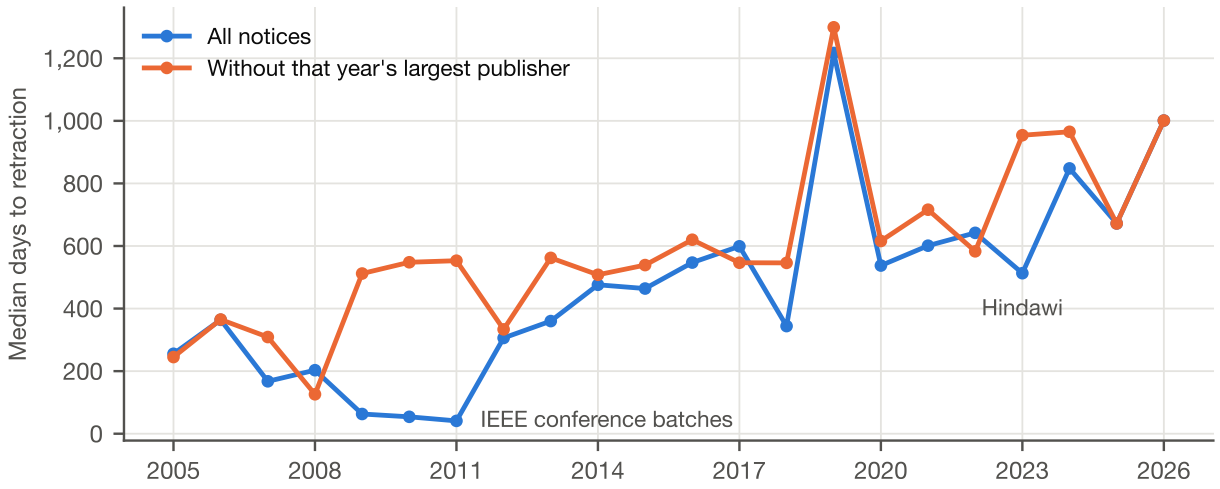


Figure 3. Median days from publication to retraction by notice year.

data are collected". What it adds is a measure of how different, and a split of the difference into its causes.

Six practices would make retraction studies reproducible.

1. **Cite the snapshot.** Report the snapshot date and the Crossref commit that produced the file, not "Retraction Watch, accessed in 2025".
2. **Archive the file.** The CSV is about 68 MB; deposit the exact copy used.
3. **Match reasons on a versioned taxonomy.** Map old labels to new before counting, or a query for "Fake Peer Review" returns zero after January 2026.
4. **Treat recent years as provisional.** In September 2024 the record of 2023 was about two thirds complete. A study should report how its counts for the last two years have moved across recent snapshots.
5. **Filter by nature before counting.** Retraction Watch holds expressions of concern, corrections, and reinstatements beside retractions; counting all of them as retractions is the error behind the 2026 correction of Salami et al. (2024a).
6. **Filter notices before using OpenAlex's flag.** Half of what it marks retracted is a notice, and it flags nine in ten notice DOIs.

6. Limitations

- Snapshots are monthly before 2026-09-29, so an edit made and reverted within a month is invisible, and the history cannot be extended before 2024-09-02, where the public repository begins.
- Identity resolution is one hop. A paper linked only through a chain of mappings appears twice, which can undercount agreement but never overcounts it.
- OpenAlex and PubMed were fetched as positive sets, so "not flagged" there mixes "known and fine" with "unknown to that source".
- PMC coverage is targeted at the 43,616 articles other registries mention, not every PMC article.

- Citation counts are per year and start in 2012.
- The hand audit has one labeller, so agreement between labellers is not measured, and its blinding is partial (Section 3.5). Three clarifications of the codebook were made during labelling; each is logged with its date.

7. Data and code availability

Code, run reports, the audit codebook, and the labelled sheet: https://github.com/TanayPratapSingh/afterlife. Findings in this draft come from runs/20261027T005839Z-findings.json; the audit sample from runs/20261027T004212Z-audit-sample.json; the audit scores from runs/20261027T221642Z-audit-score.json. `afterlife findings` rebuilds every number from the landed snapshots. The 39 Retraction Watch snapshots and every other source file, exactly as fetched, with checksums, are archived at https://doi.org/10.17605/OSF.IO/D7GSF.

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